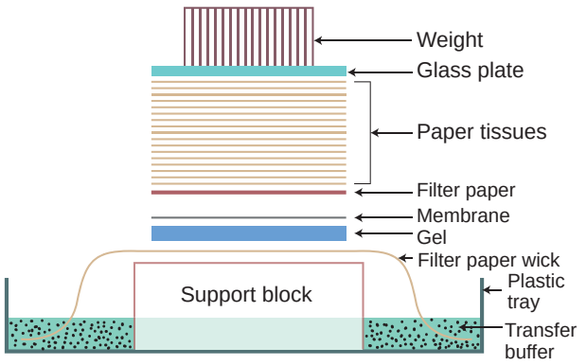


Figure 4.21: Diagrammatic representation of a typical blotting apparatus

Northern Blot

It was found that RNA is not binding to cellulose nitrate. Therefore, Alwin *et al.* (1979) devised a procedure in which RNA bands are transferred from the agarose gel into nitrocellulose filter paper. This transfer of RNA from gel to special filter paper is called Northern Blot hybridization. The filter paper used for Northern blot is Amino Benzyloxymethyl Paper which can be prepared from Whatman 540 paper.

Western Blot

Refers to the electrophoretic transfer of proteins to blotting papers. Nitrocellulose filter paper can be used for western blot technique. A particular protein is then identified by probing the blot with a radio-labelled antibody which binds on the specific protein to which the antibody was prepared.

4.7.6 Bioassay for Target Gene Effect

Target gene is target DNA, foreign DNA, passenger DNA, exogenous DNA, gene of interest or insert DNA that is to be either cloned or specifically mutated. Gene targeting experiments have been targeting the nuclei and this leads to 'gene knock-out'. For this purpose, two types of targeting vectors are used. They are insertion vectors and replacement or transplacement vectors.

1. Insertion vectors are entirely inserted into targeted locus as the vectors are linearized within the homology region. Initially, these vectors are circular but during insertion, become linear. It leads to duplication of sequences adjacent to selectable markers.

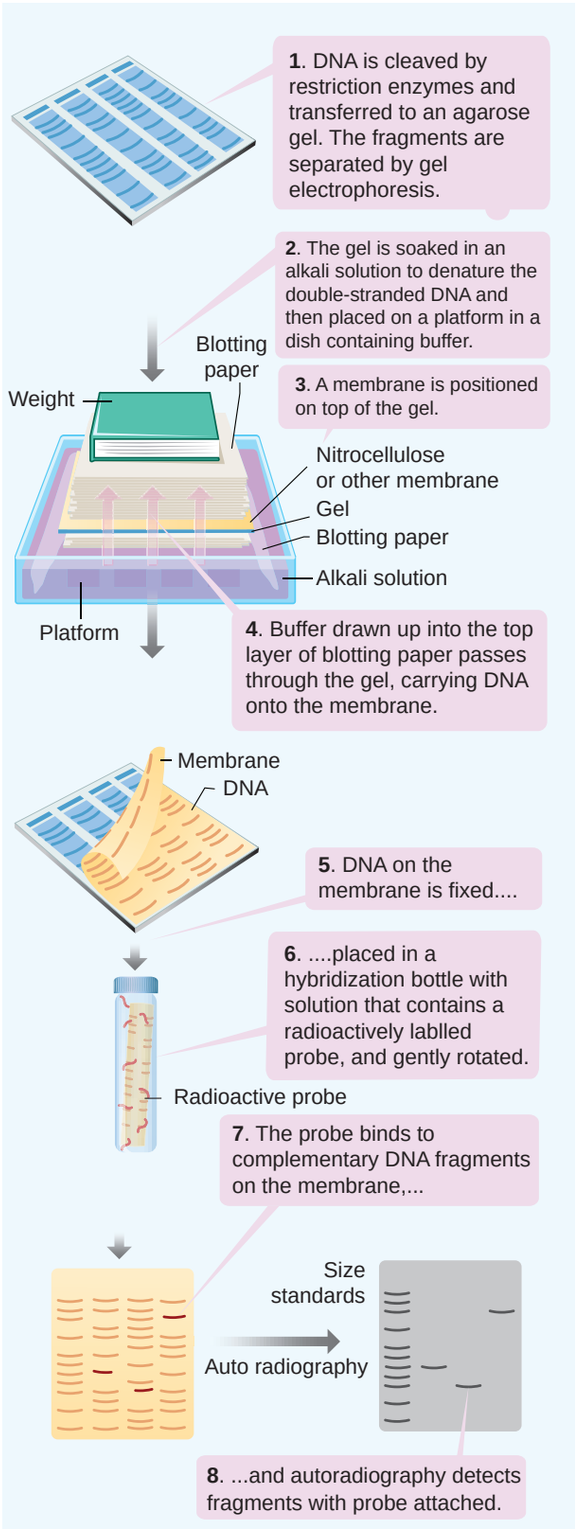


Figure 4.22: Steps involved in southern blotting technique

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